Project Proposal

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EN-2160 : Engineering Design Realization

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1 Problem Statement

1.1 Introduction

In Sri Lanka the main sources of water supply to public is provided through sources such as piped water, public taps, tube wells, protected wells, semi protected wells, rural water supply projects, bottled water Then in case of pipe water distribution the main body that is responsible for that is National Water Supply and Drainage Board(NWSDB). About 75% of urban population and about 40% of the rural population are supplied water through this distribution newtorks. Main resovioursfor them are tube wells and surfae water resoviours(eg. lakes, rivers etc)
Currently, meters are placed at the households or more generally at the user end to monitor the amount of usage for management purposes such as billing.

1.2 Current difficulties

- **Manual measurement taking**
  The main thing we observe here is that water meter monitoring is something that has the potential to move towards automated monitoring to improve the quality of service by minimizing the errors occur and minimizing the costs by reducing the human interactions.

- **Real-time monitoring and accessing usage history**
  In the current system, there is a lack of accessability to realtime data and to see the usage history. The only method that is possible right now is to look it from the meter directly or get annual data from the bill.

2 Proposed Solution

2.1 Introduction

We thought through the problem and decided to introduce an integrated device with a smart meter and a smart ball-valve.

2.2 Functionality

Here, using our proposed device we intend to let user access real-tine water usage and to make it convenient to analyze past data by providing an application for better user experience. In addition to that from the side of the water board we can illuminate the necessity of checking the meters by visiting house holds.

2.3 Existing products

In the market we can find the smart meter and the ball valve separately only in the international market. If someone to import them that will cost a huge amount considering all the expenditures with taxes and all.

2.4 Stake-holders

Some of the stake-holders that we identified regarding our project are,

- House-holds
- Farmers
- Water-board

Resources

- https://www.statistics.gov.lk/Resource/en/Health/DemographicAndHealthSurveyRepo
  2016-Chapter2.pdf
- https://drive.google.com/file/d/1m3dZbix2ck5f2yChTXGhR04r8Bdc5cNK/view